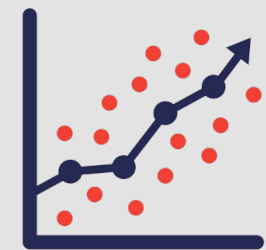


วิเคราะห์ข้อมูลด้วย RapidMiner

Asst.Prof. Dr.Damrongrit Setsirichok

Faculty of Digital Technology

Chitralada Technology Institute



หัวข้อ

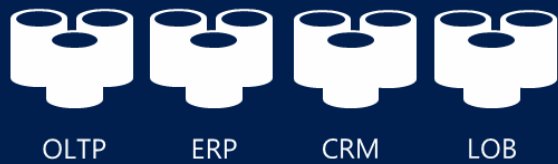
- แนะนำซอฟต์แวร์ RapidMiner
- การนำข้อมูลเข้าและจัดการข้อมูลเบื้องต้นบนซอฟต์แวร์ RapidMiner
- การสร้างแบบจำลองข้อมูลบนซอฟต์แวร์ RapidMiner เบื้องต้น

กระบวนการ และ เครื่องมือ



Pipeline

Source Systems



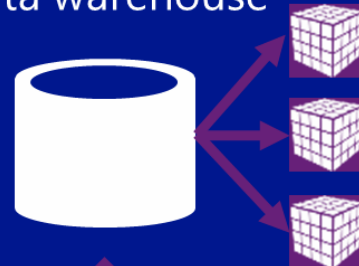
New Data



ETL



Data warehouse



BI & analytics



Copy Data



Hadoop



Data Science



Magic Quadrant for Data Science and Machine Learning Platforms

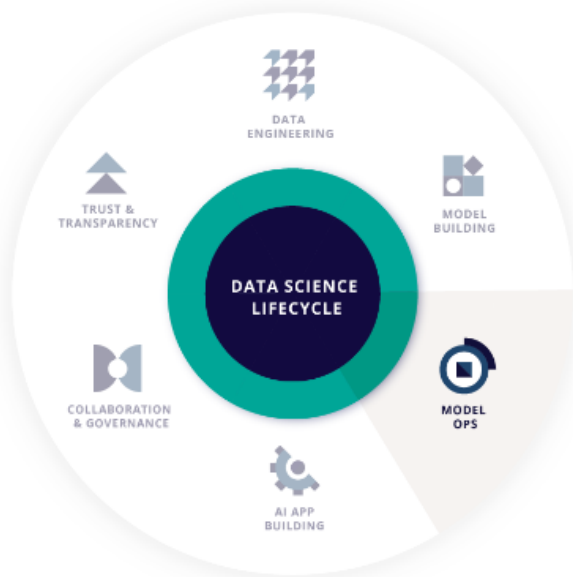


Figure 1: Magic Quadrant for Data Science and Machine Learning Platforms



RapidMiner

Data Analytics and AI Platform



<https://altair.com/altair-rapidminer>



Data Engineering



Model Building



ML Ops

Simplify operations, no matter how complex models are, or how they were created. Deploy, evaluate, compare, monitor, manage and swap any model.

[Learn more >](#)



AI App Building



Collaboration & Governance



Trust & Transparency

Instill confidence in your work with explainable AI. Understand how models are making predictions and expose downstream ROI.

RapidMiner Platform

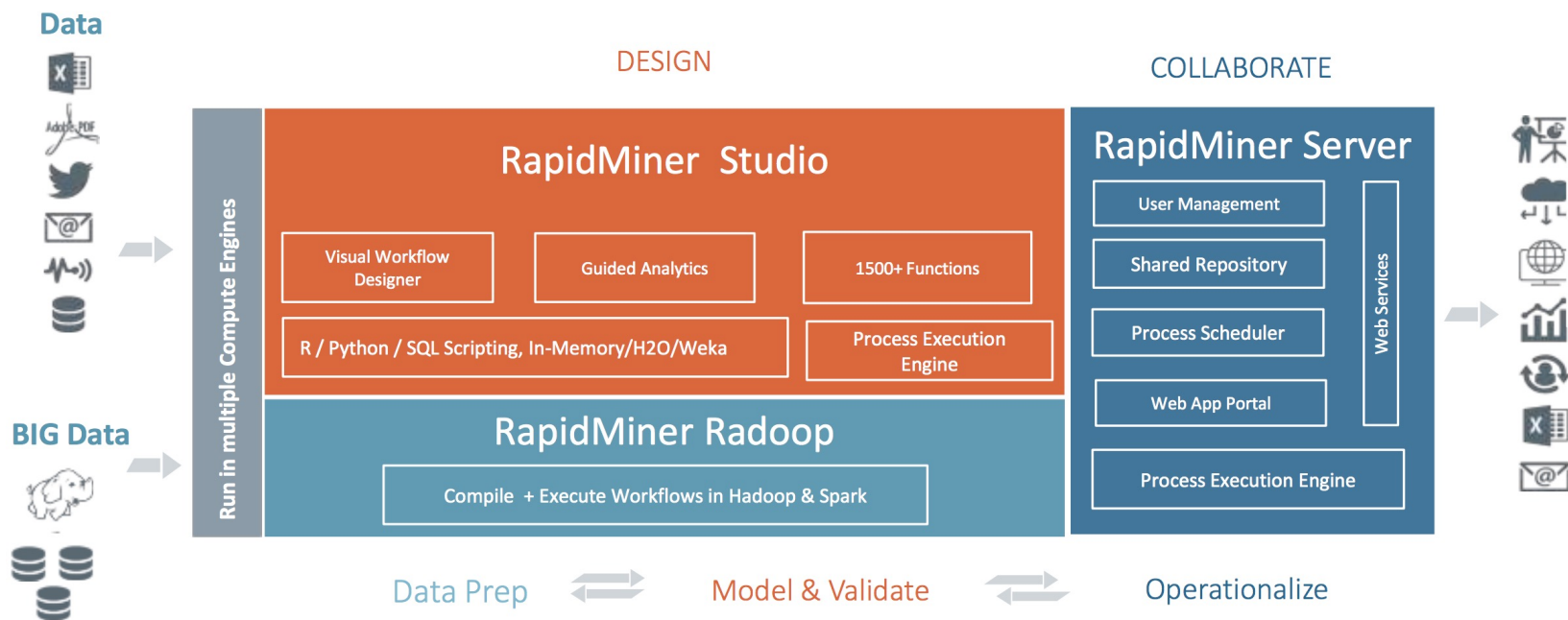
<https://rapidminer.com>



Unified Data Science Platform



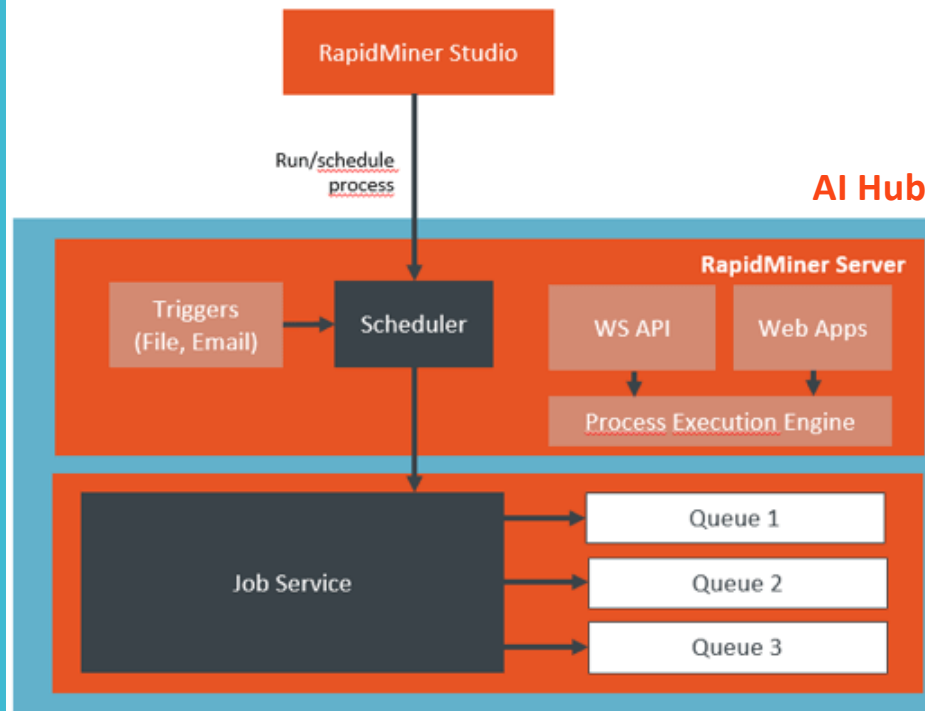
RAPIDMINER



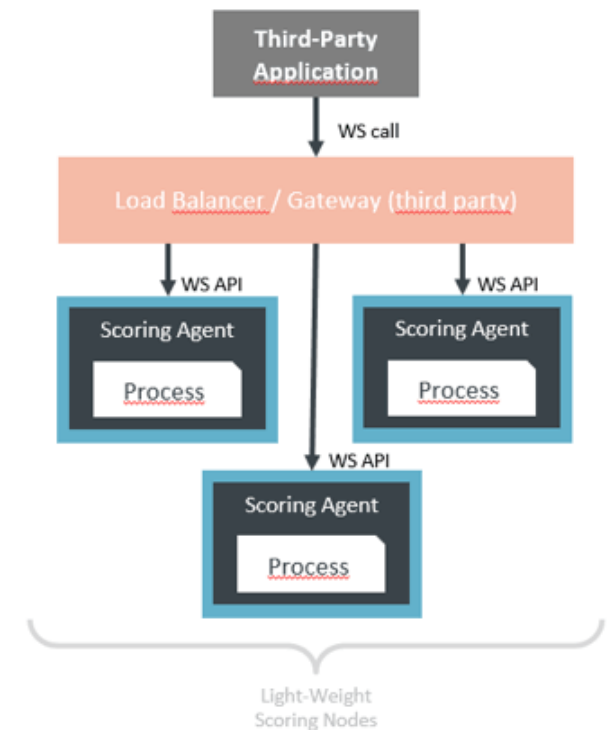
RapidMiner Platform

<https://rapidminer.com>

Development - Training



Operationalization - Scoring



RapidMiner Studio

<https://altair.com/academic-program>

RapidMiner Studio Pricing

Visual workflow designer for the entire analytics team

Free	Professional	Enterprise
\$0	\$7,500 PER USER, PER YEAR	\$15,000 PER USER, PER YEAR
10,000 Data Rows 1 Logical Processor Community Support 30 Day Trial of Enterprise	100,000 Data Rows 2 Logical Processors Enterprise Support Turbo Prep Auto Model	Unlimited Data Rows Unlimited Logical Processors Enterprise Support Turbo Prep Auto Model Automated Model Ops Background Process Execution
DOWNLOAD NOW	CONTACT SALES	CONTACT SALES

RapidMiner Studio

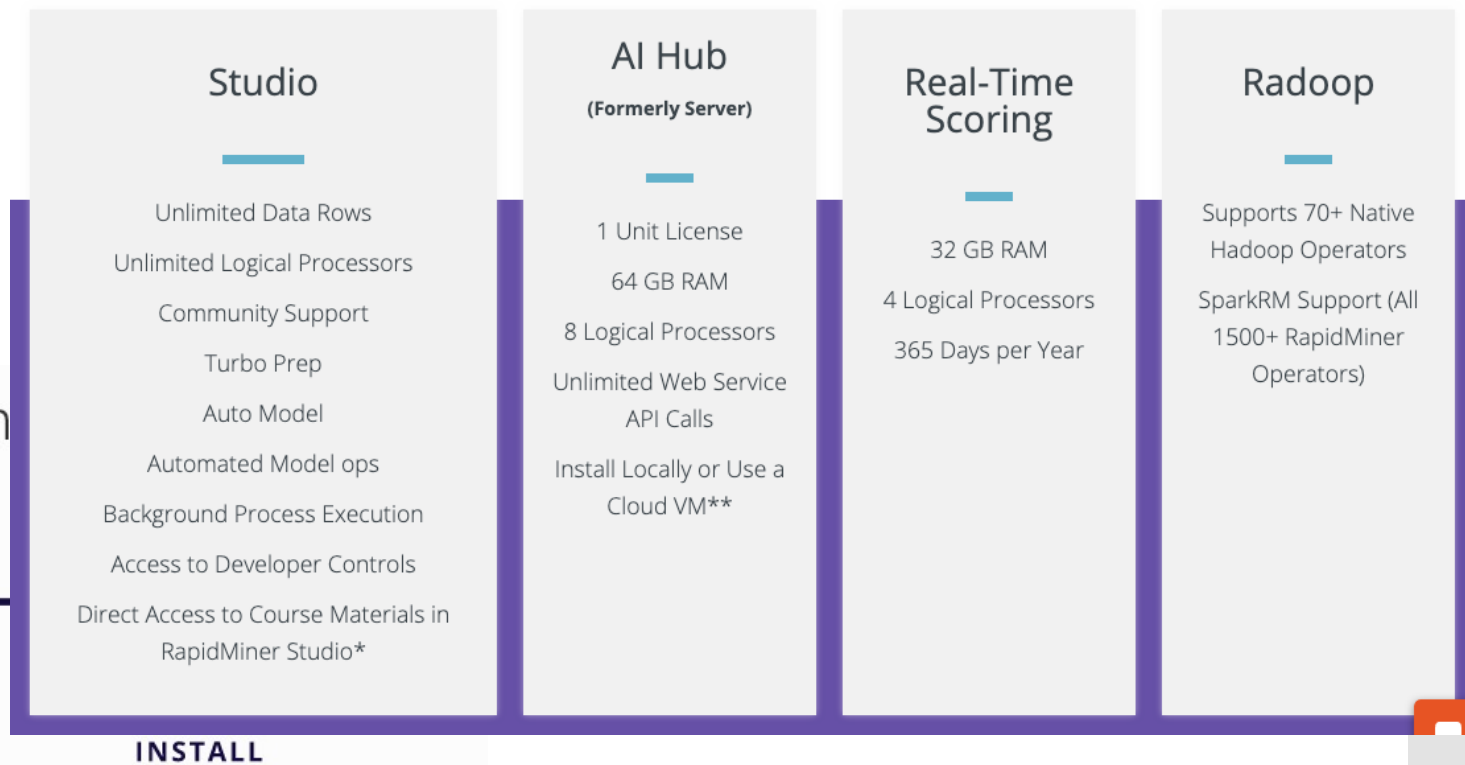
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Eligibility for Educational License

RapidMiner's Educational License is **available** for:

- students currently enrolled in an academic institution
- educators teaching at an academic institution

Note that you must have a valid academic email address in order to apply for the Educational License.

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RapidMiner's Educational License is **not available** for commercial, non-profit organizations, or funded research.

<https://docs.rapidminer.com/latest/studio/installation/license/rapidminer.html>

RapidMiner Academy

→ academy.rapidminer.com

RAPIDMINER ACADEMY

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Jump right into our first Learning Path



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All Courses, Paths and Micro-Learning elements



CERTIFICATION

The best place for your career Advancement



USE CASE GOLDMINE

Examples are the best way to learn Applied Data Science

RapidMiner Community

 **RAPIDMINER** | An  **ALTAIR** Company

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Best Of...

Unanswered 100+

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Server 782 Errors 719

Getting Started 668

Developer 598 Clustering 519

Java 466 Extensions 432

Highest Rated Posts



Ihlilcken
November 2021
Panda Module detected but unable to load (MacOS)
I am trying to link my Python 3.10 with RapidMiner but always get this message. Does anyone know how to



MartinLiebig
July 2020
Re: [New Extension] How to do Time Series Forecasting with just 1 Operator!



MartinLiebig
January 2021
New Extension: Interpretations - SHAP, LIME and Shapely
Dear Community I am happy to announce that @pschlunder and I

What are you using Altair RapidMiner for?

- ☐ Commercial purposes (e.g., business, evaluation, not-for-profit)
☒ Educational purposes (e.g., educator, student)

First name:

Last name:

Country:

University:

Role:

Email address:

Create a password:

<https://my.rapidminer.com/nexus/account/index.html#signup>

สมัคร Educational Account โดยกรอกข้อมูลตามจริง
และ email ของสถาบันฯ



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Licenses

View license keys for your Altair RapidMiner products.

AI Studio

AI Hub

Scoring Agent

Radoop

Go (Legacy)

▼ AI Studio 7.2+

AI Studio

Educational

Damrongrit Setsirichok

🕒 Expires **Thu, Apr 9th 2026 (7 months left)**

View License Key

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ประเภท Educational



Downloads

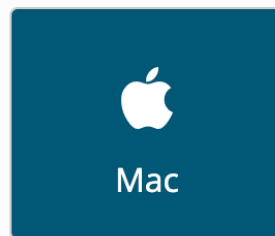
Click on an Altair AI product of your choice to download it.

AI Studio 2025.1

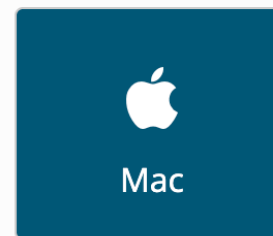
Click on your operating system to start the download:



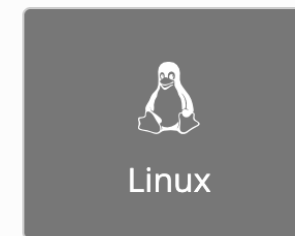
64bit



x64



ARM

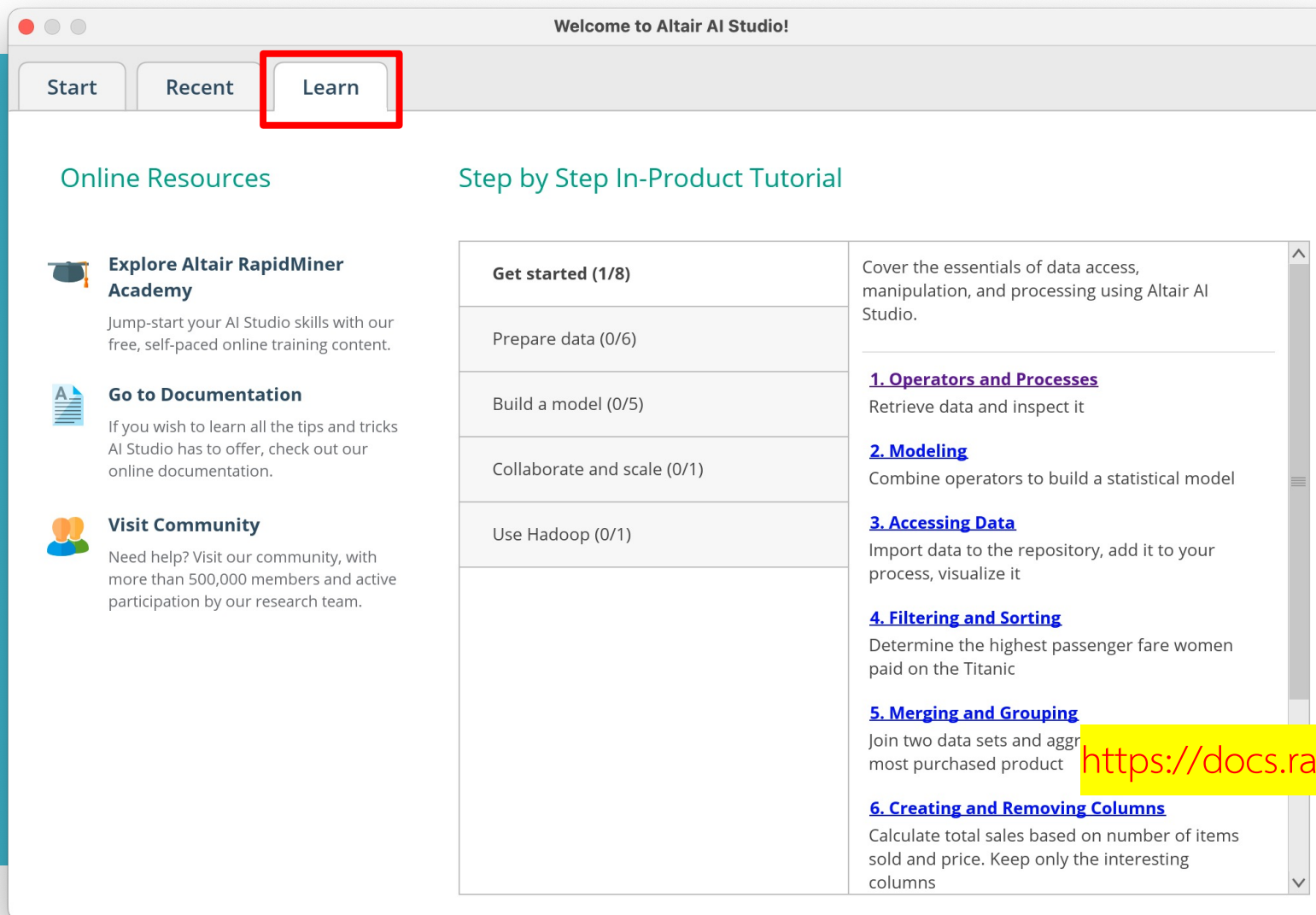


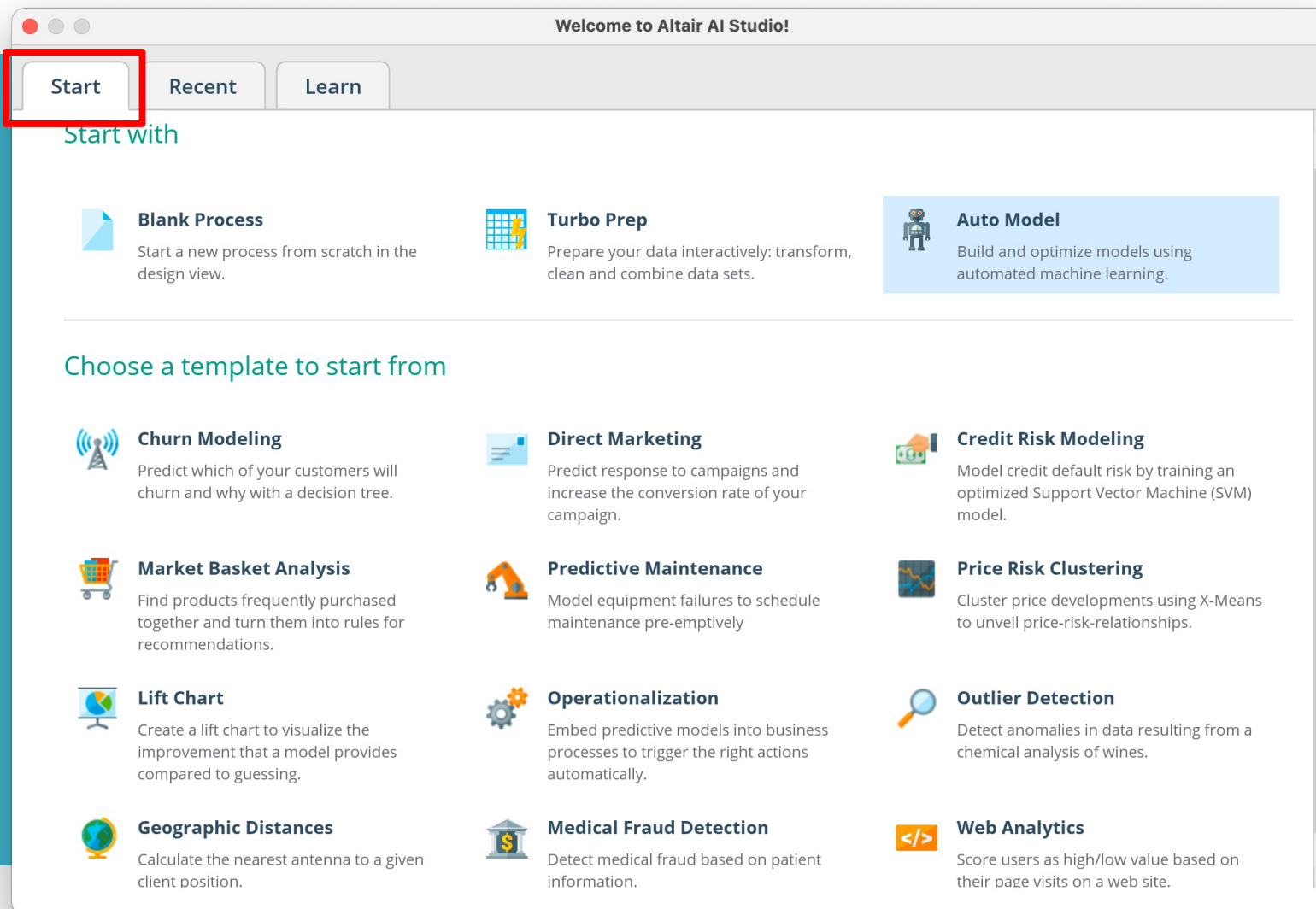
64bit
Requires: [Java 17](#)

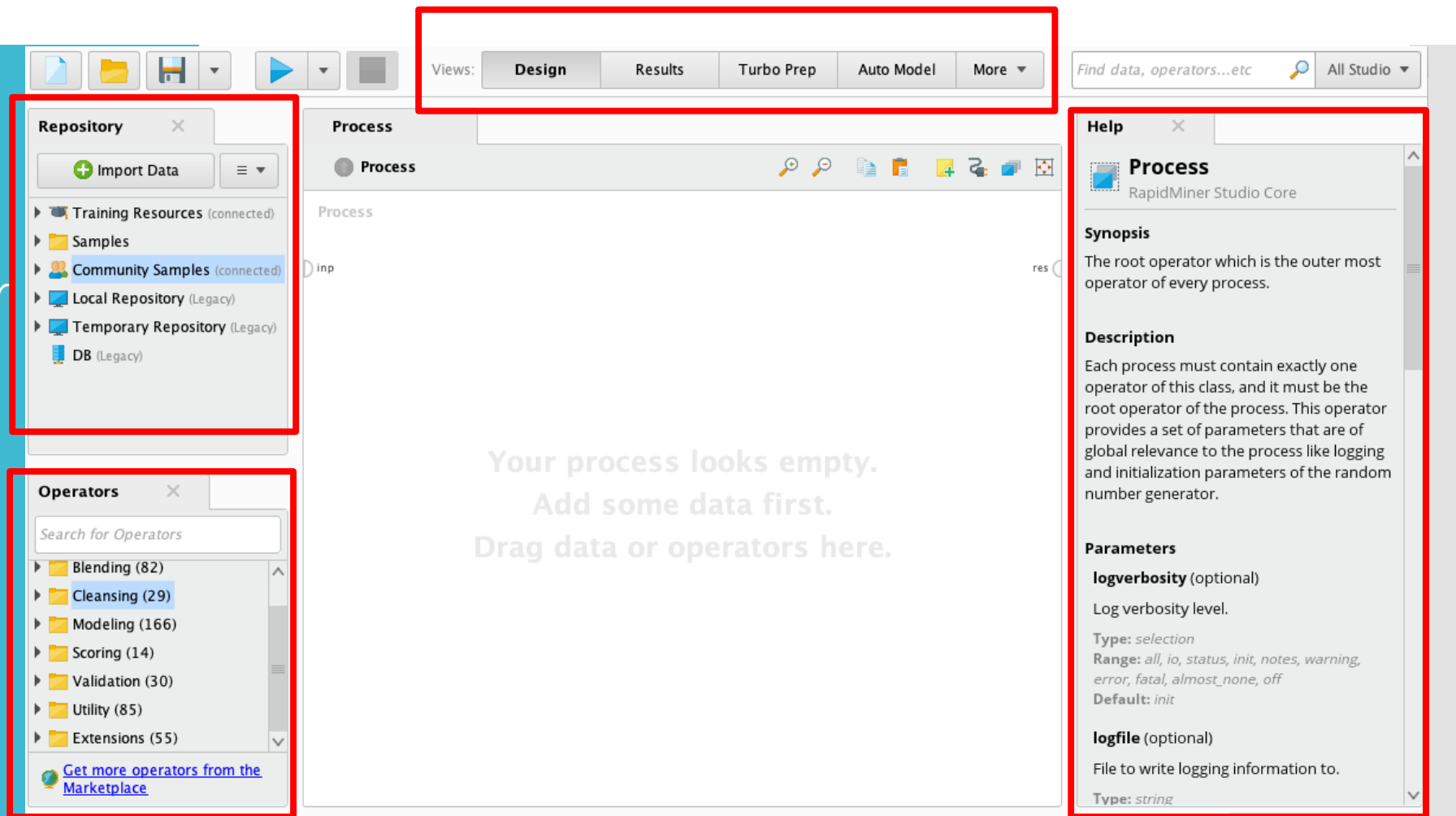
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The logo for RapidMiner Studio Design is displayed on a solid teal background. The text "RapidMiner" is in a white sans-serif font, with "Studio" below it in a larger, lighter weight. "Design" is positioned further down in a smaller, regular weight font. A thin red rectangular border is visible on the right side of the image.The logo for RapidMiner Studio Design is displayed on a solid teal background. The text "RapidMiner" is in a white, sans-serif font, with the "i" in "Miner" being lowercase. Below it, the word "Studio" is also in a white, sans-serif font. At the bottom, the word "Design" is in a white, sans-serif font. To the right of the text, there are two vertical red rectangular outlines, one above the other, which are part of the logo's design.

RapidMiner Studio

Results

Views: Design Results Turbo Prep Auto Model Deploy

Result History

Deep Learning Model (Deep Learning) PerformanceVector (Performance)

Criterion

- accuracy
- precision
- recall
- AUC (optimistic)
- AUC
- AUC (pessimistic)

Performance

Description

Table View Plot View

accuracy: 97.67%

	true ham	true spam	class prec
pred. ham	475	6	98.75%
pred. spam	7	69	90.79%
class recall	98.55%	92.00%	

RapidMiner Studio

Turbo prep

The screenshot displays the RapidMiner Studio interface in the Turbo Prep tab. The top navigation bar includes icons for file operations and tabs for Design, Results, and Turbo Prep. A search bar and 'All Studio' dropdown are also present. The left sidebar, titled 'Data Sets', features a '+ LOAD DATA' button and details for the 'Iris' dataset: 'Created by Retrieve Iris', 'Rows: 150', 'Columns: 6', and 'Last Change: None'. The main workspace shows the 'Iris' dataset with a toolbar for TRANSFORM, CLEANSE, GENERATE, PIVOT, and MERGE. Above the data table are histograms for each column. The data table has six columns: a1 (Number), a2 (Number), a3 (Number), a4 (Number), id (Category), and label (Category). The first six rows of data are visible, showing various numerical values and the 'Iris-setosa' label.

a1 Number	a2 Number	a3 Number	a4 Number	id Category	label Category
5.100	3.500	1.400	0.200	id_1	Iris-setosa
4.900	3	1.400	0.200	id_2	Iris-setosa
4.700	3.200	1.300	0.200	id_3	Iris-setosa
4.600	3.100	1.500	0.200	id_4	Iris-setosa
5	3.600	1.400	0.200	id_5	Iris-setosa
5.400	3.900	1.700	0.400	id_6	Iris-setosa

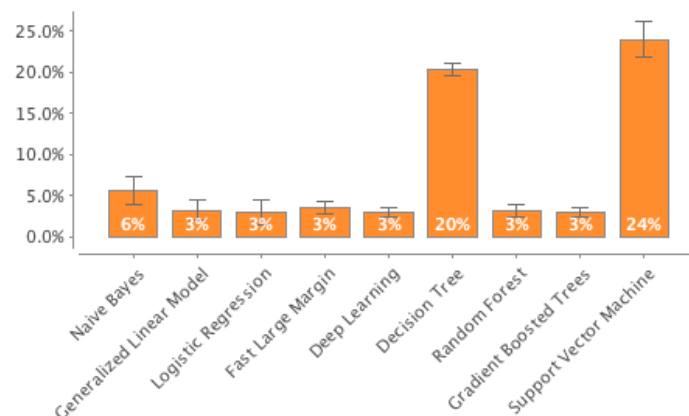
RapidMiner Studio

Auto model

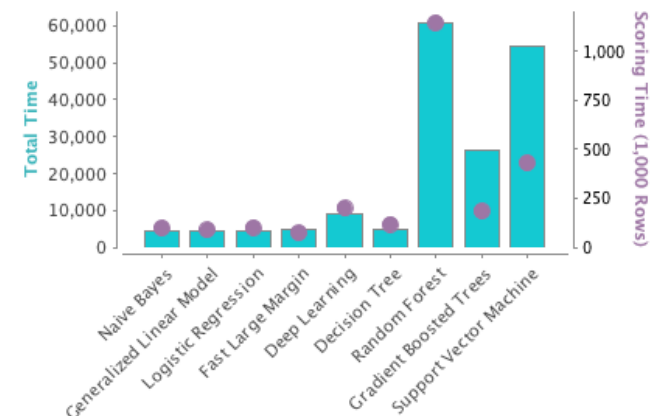
Overview

Number of Models: 205

Classification Error



Runtimes (ms)



Classification Error ▼

Model		Classification Error	Standard Deviation	Gains	Total Time
Naive Bayes		5.6%	± 1.7%	256	4 s
Generalized Linear Model		3.2%	± 1.2%	262	5 s
Logistic Regression		2.9%	± 1.5%	250	4 s
Fast Large Margin		3.5%	± 0.7%	264	5 s
Deep Learning		2.9%	± 0.6%	276	9 s
Decision Tree		20.3%	± 0.7%	118	5 s
Random Forest		3.2%	± 0.7%	274	1 min 0 s
Gradient Boosted Trees		2.9%	± 0.6%	276	26 s
Support Vector Machine		24.0%	± 2.1%	110	54 s

RapidMiner
Studio

Auto model

Views: Design Results Turbo Prep **Auto Model** Deployments Find data, operators...etc All Studio

Auto Model

Load Data Select Task Prepare Target Select Inputs Model Types **Results**

RESTART BACK OPEN PROCESS EXPORT **DEPLOY**

Results

- Comparison
 - Overview
 - ROC Comparison
- Naive Bayes
 - Model
 - Weights
 - Simulator
 - Performance
 - Lift Chart
 - Predictions
 - Production Model
- Generalized Linear Model
- Logistic Regression
 - Model
 - Weights
 - Simulator
 - Performance**
 - Lift Chart
 - Predictions

Logistic Regression – Performance

Performances

Criterion	Value	Standard Deviation
Accuracy	97.1%	± 1.5%
Classification Error	2.9%	± 1.5%
AUC	99.0%	± 1.0%
Precision	100.0%	± 0.0%
Recall	91.7%	± 4.9%
F Measure	95.6%	± 2.7%
Sensitivity	91.7%	± 4.9%
Specificity	100.0%	± 0.0%

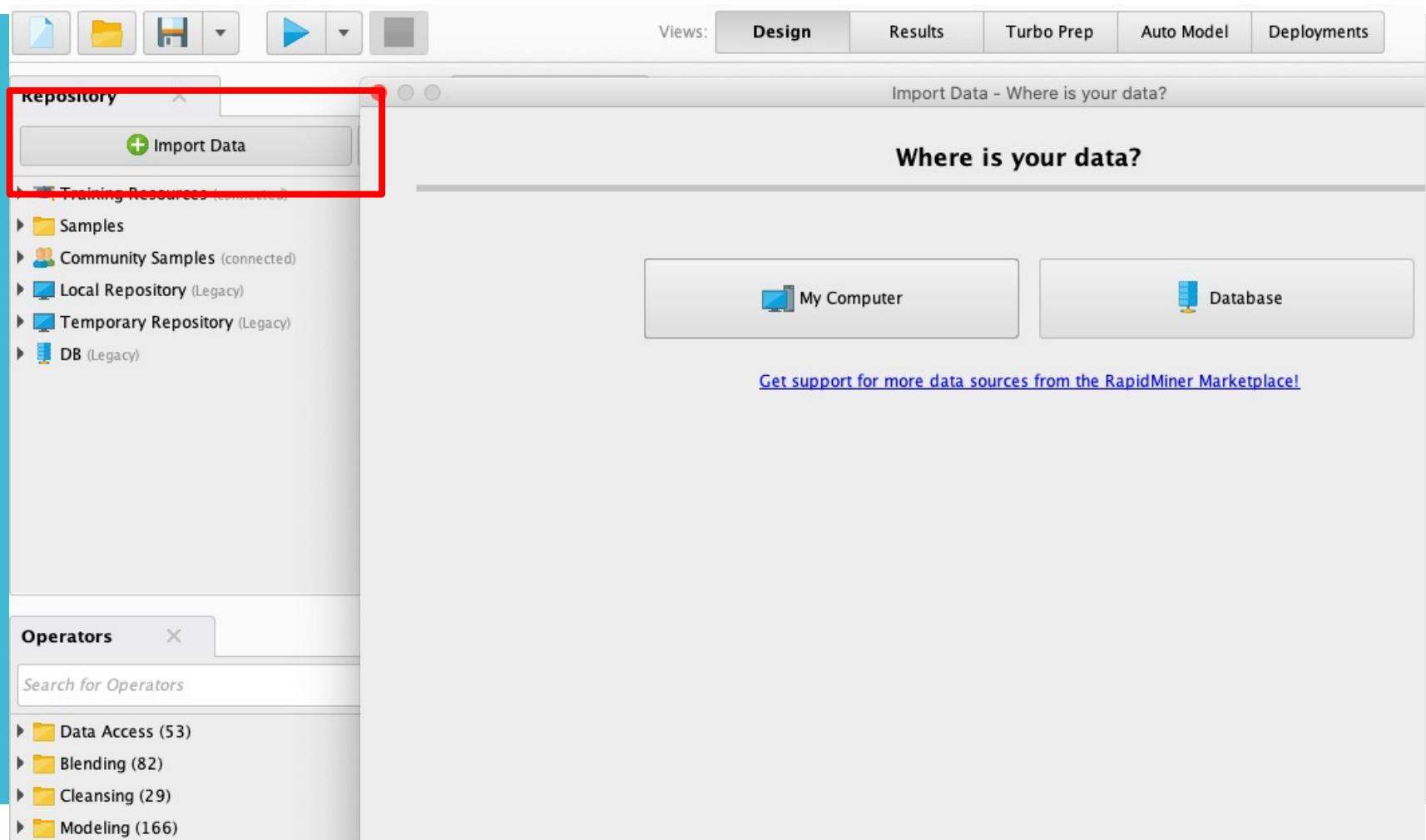
Confusion Matrix

	true No	true Yes	class precision
pred. No	238	11	95.58%
pred. Yes	0	125	100.00%
class recall	100.00%	91.91%	

SAVE RESULTS

RapidMiner
Studio

Import Data



Dataset
(ชุดข้อมูล)

Sample

Attribute/ Feature

Label

Outlook	Temperature	Humidity	Windy	Play
sunny	85	85	FALSE	no
sunny	80	90	TRUE	no
overcast	83	78	FALSE	yes
rain	70	96	FALSE	yes
rain	68	80	FALSE	yes
rain	65	70	TRUE	no
overcast	64	65	TRUE	yes
sunny	72	95	FALSE	no
sunny	69	70	FALSE	yes
rain	75	80	FALSE	yes
sunny	75	70	TRUE	yes
overcast	72	90	TRUE	yes
overcast	81	75	FALSE	yes
rain	71	80	TRUE	no

Example

Attribute Type

Attribute Type		Description	Examples
Categorical (Qualitative)	Nominal	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another. (=, $\neq$)	zip codes, employee ID numbers, eye color, gender
	Ordinal	The values of an ordinal attribute provide enough information to order objects. ($<$, $>$)	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers
Numeric (Quantitative)	Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. (+, -)	calendar dates, temperature in Celsius or Fahrenheit
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current

RapidMiner Studio

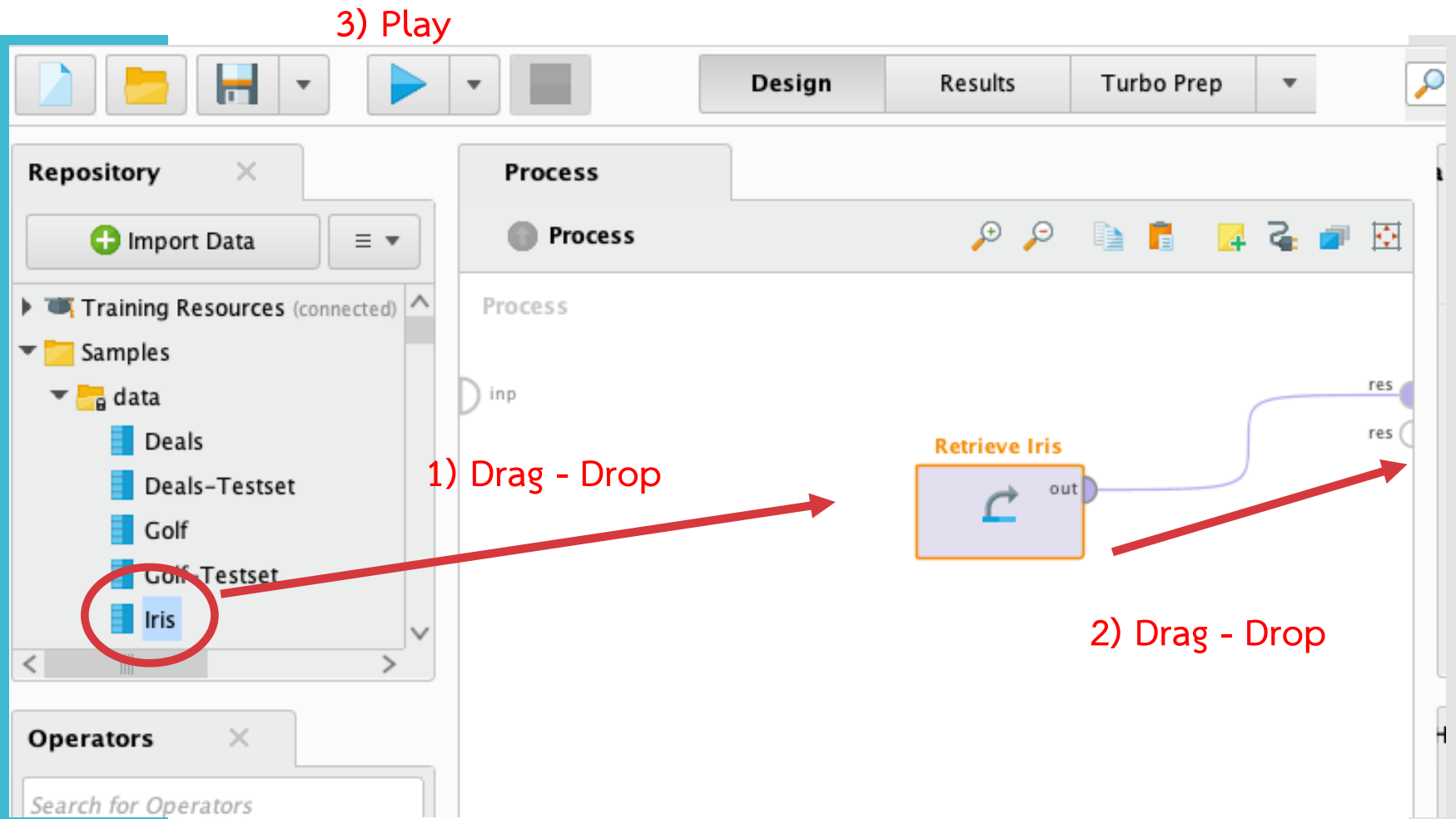
Overview

- Data Exploration
- Visualizing Data In RapidMiner
- Turbo Prep



RapidMiner Studio

Data Exploration & Visualizing



Views:

Design

Results

Turbo Prep

More ▾

Find data, operator:

Result History

ExampleSet (Retrieve Iris) ×

ExampleSet (//Local Repository/data/rul_hrs) ×

Data

Statistics

Visualizations

Annotations

Open in

Turbo Prep
 Auto Model

Filter (150 / 150 examples):

all ▾

Row No.	id	label	a1	a2	a3	a4
1	id_1	Iris-setosa	5.100	3.500	1.400	0.200
2	id_2	Iris-setosa	4.900	3	1.400	0.200
3	id_3	Iris-setosa	4.700	3.200	1.300	0.200
4	id_4	Iris-setosa	4.600	3.100	1.500	0.200
5	id_5	Iris-setosa	5	3.600	1.400	0.200
6	id_6	Iris-setosa	5.400	3.900	1.700	0.400
7	id_7	Iris-setosa	4.600	3.400	1.400	0.300
8	id_8	Iris-setosa	5	3.400	1.500	0.200
9	id_9	Iris-setosa	4.400	2.900	1.400	0.200
10	id_10	Iris-setosa	4.900	3.100	1.500	0.100
...	...	...	...	...	...	...

Views:

Design

Results

Turbo Prep

More

Find

Result History

ExampleSet (Retrieve Iris)

Data

Statistics

Visualizations

Annotations

Name	Type	Missing	Filter (6 / 6 attributes):	
Id id	Nominal	0	Least id_99 (1)	Most id_1 (1)
Label label	Nominal	0	Least Iris-virginica (50)	Most Iris-setosa (50)
a1	Real	0	Min 4.300	Max 7.900
a2	Real	0	Min 2	Max 4.400
a3	Real	0	Min 1	Max 6.900
a4	Real	0	Min 0.100	Max 2.500

Type

Polynomial

Polynomial

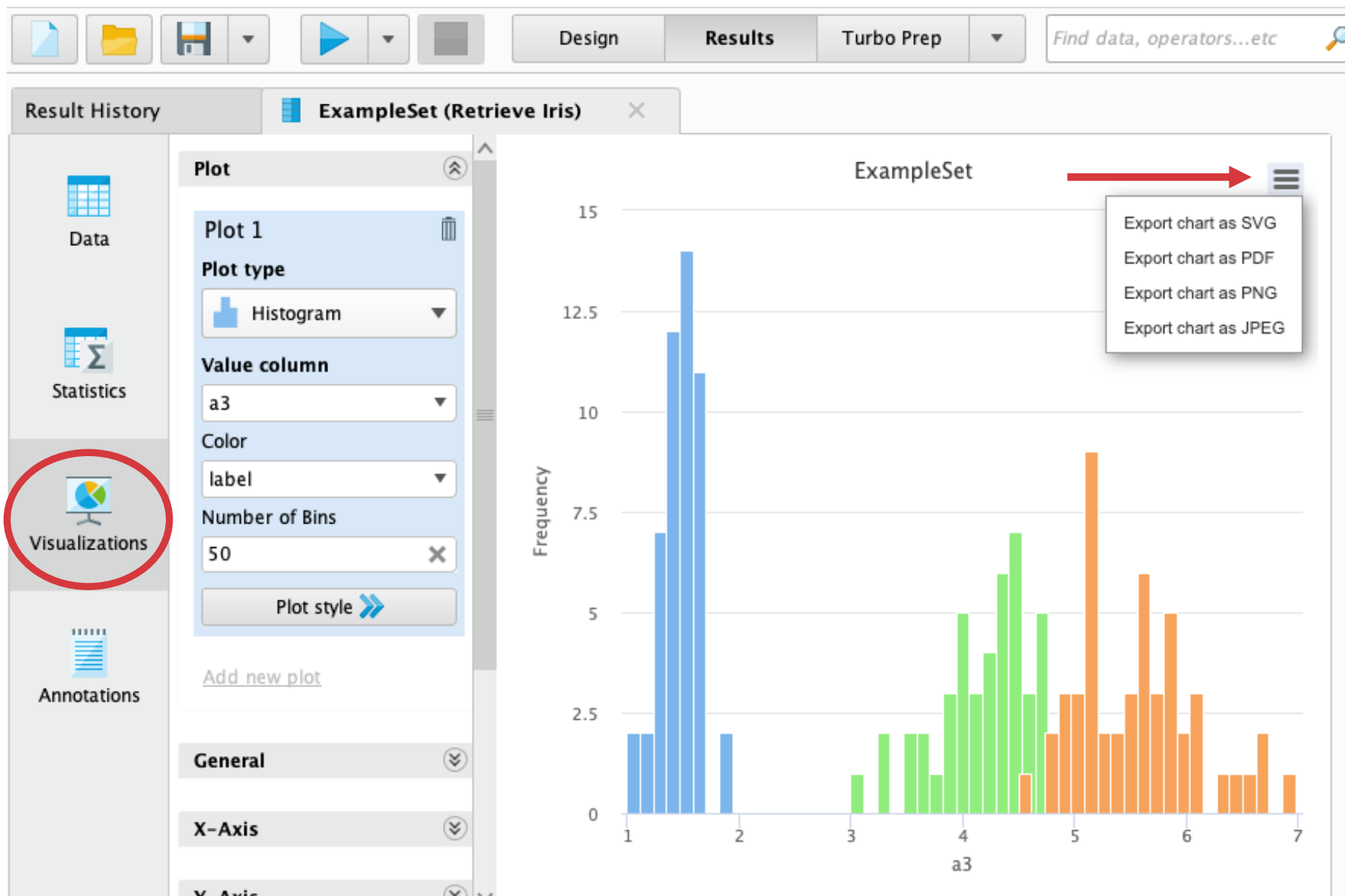
Binominal

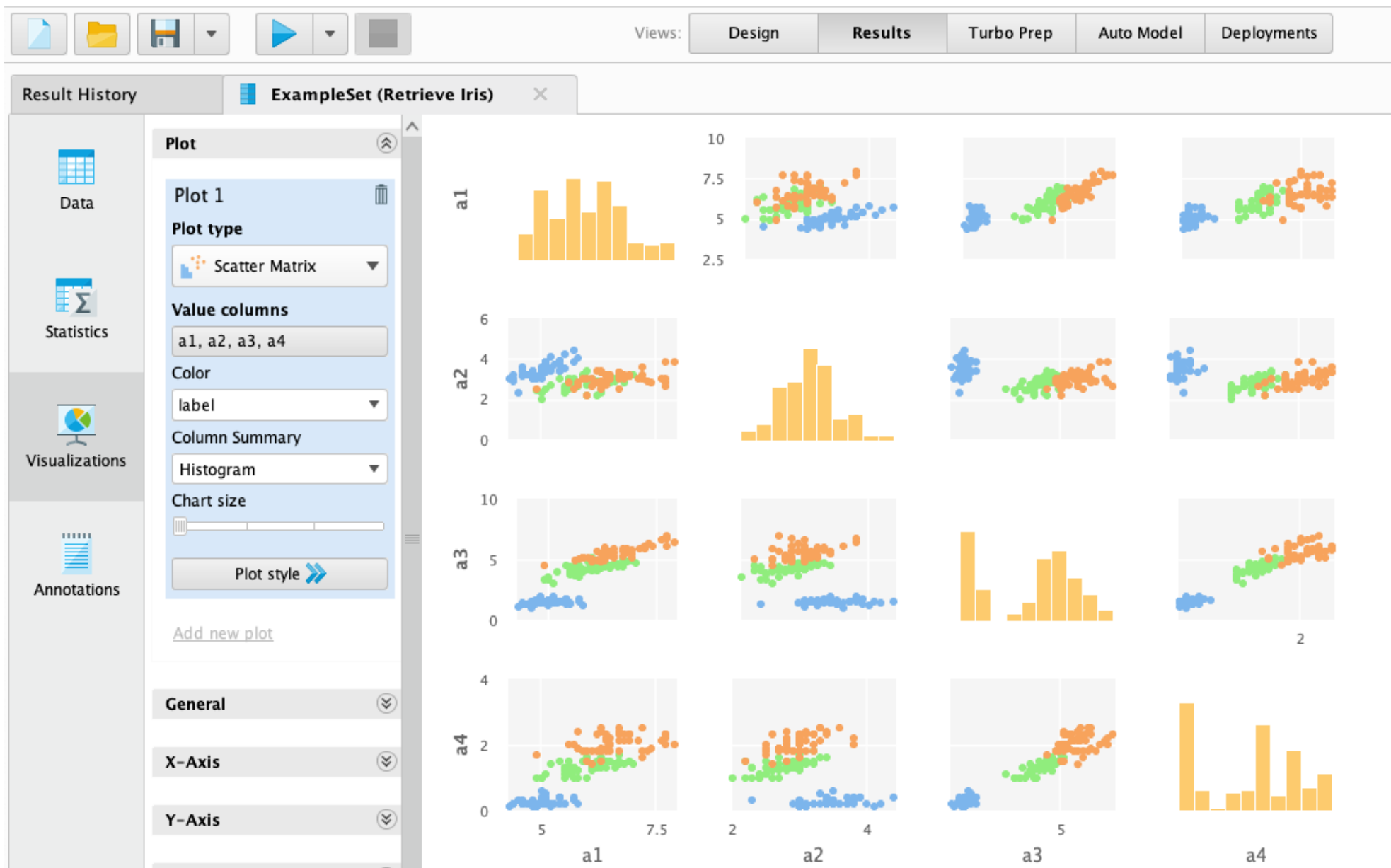
Real

Integer

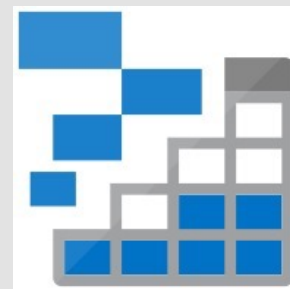
Integer

Polynomial





Turbo Prep



RapidMiner Studio

Turbo Prep - Transform

Iris

Add new data sets on the left. Details for the selected data are shown below. You can change the data with the following action

 **TRANSFORM**

 CLEANSE

 GENERATE

 PIVOT

 MERGE

Transform your data with actions like filtering, removal of columns, and much more.

a1 Number	a2 Number
5.100	3.500
4.900	3
4.700	3.200

Turbo Prep

Transform

1 column selected

RENAME

CHANGE TYPE

REMOVE

COPY

FILTER

RANGE

SAMPLE

SORT

REPLACE

SPLIT

Iris

Select columns to transform (hold Shift for selecting a range)

 COMMIT TRANSFORMATION CANCEL

a1 Number	a2 Number
5.100	3.500
4.900	3
4.700	3.200
4.600	3.100
5	3.600
5.400	3.900
4.600	3.400
5	3.400
4.400	2.900
4.900	3.100
5.400	3.700
4.200	3.100

RapidMiner Studio

Turbo Prep - Cleanse

Iris

Add new data sets on the left. Details for the selected

✂ TRANSFORM **CLEANSE** 📊 GENERATE

Prepare your data for k

a1 Number	a2 Number
5.100	3.500
4.900	3

Turbo Prep

Cleanse

1 column selected

COMMIT CLEANSE

AUTO CLEANSING

REMOVE LOW QUALITY

REMOVE CORRELATED

REPLACE MISSING

NORMALIZATION

DISCRETIZATION

DUMMY ENCODING

PCA

REMOVE DUPLICATES

Iris

Select a column to clean (hold Shi

COMMIT CLEANSE

a1 Number
5.100
4.900
4.700
4.600
5
5.400
4.600
5
4.400
4.900
5.400

36

Iris

Add new data sets on the left. Details for the selected data are shown below. You can ch

TRANSFORM CLEANSE **GENERATE** PIVOT MERGE

Generate new columns for your d

Turbo Prep

Generate

Search text



label
Category

a1
Number

a2
Number

a3
Number

a4
Number

id
Category

Iris

Generate a new column below. You can type a formula or drag in columns

COMMIT GENERATE CLEAR ALL CANCEL

Name: New_class

Formula:

```
if ( [label]=="Iris-setosa", 0,  
    if ( [label]=="Iris-versicolor", 1, 2 )  
)
```

UPDATE PREVIEW

Preview:

New_class Number	label Category	a1 Number	a2 Number
0	Iris-setosa	5.100	
0	Iris-setosa	4.900	
0	Iris-setosa	4.700	
0	Iris-setosa	4.600	

Used Column Generated Column

150 rows -
7 columns
(2 nominal,
5

RapidMiner
Studio

Turbo Prep
- Generate

<https://academy.rapidminer.com/learn/video/turbo-prep-introduction>

RapidMiner
Studio

Turbo Prep
- Pivot

Turbo Prep

Pivot

Search text



a1
Number

a2
Number

a3
Number

a4
Number

id
Category

label
Category

Iris

Drag columns from the left to build a pivot table. ⓘ

Σ COMMIT PIVOT

CLEAR TABLE

CANCEL

Group-By

Column Grouping

+

Drop a column on the **left box** to use its values as row IDs.
Drop a column on the **top box** to use its values as column headers.
Columns dropped on the **bottom box** will be shown aggregated in the table.

+

Aggregates

+

RapidMiner Studio

Turbo Prep - Pivot

Turbo Prep

Pivot



a1
Number

a2
Number

a3
Number

a4
Number

id
Category

label
Category

Iris

Drag columns from the left to build a pivot table. ⓘ

Σ COMMIT PIVOT

CLEAR TABLE

CANCEL

Group-By

label

Column Grouping

+

label

Iris-setosa

Iris-versicolor

Iris-virginica

a1
average

5.006

5.936

6.588

+

■ 1 Group-by Column

■ 1 Aggregate

3 rows - 2 colu

Aggregates

a1
average

RapidMiner Studio

Turbo Prep - Merge

Turbo Prep

Merge

Merge With

Select a second data set...

Select a second data set...

100% class_iris

Merge Type

Select a data set first...

Select a data set first...

Inner Join

Outer Join

Left Join

Right Join

Append

Join Keys ⓘ

☐ Use Row Numbers as Keys

+ ADD JOIN KEYS

label

label

Iris

Select a second data set to merge with on the left. Define how both data sets should be merged and see a previ.

COMMIT MERGE

CANCEL

UPDATE PREVIEW

a1 Number	a2 Number	a3 Number	a4 Number	id Category	label Category
5.100	3.500	1.400	0.200	id_1	Iris-setosa
4.900	3	1.400	0.200	id_2	Iris-setosa
4.700	3.200	1.300	0.200	id_3	Iris-setosa
4.600	3.100	1.500	0.200	id_4	Iris-setosa
5	3.600	1.400	0.200	id_5	Iris-setosa
5.400	3.900	1.700	0.400	id_6	Iris-setosa
4.600	3.400	1.400	0.300	id_7	Iris-setosa
5	3.400	1.500	0.200	id_8	Iris-setosa
4.400	2.900	1.400	0.200	id_9	Iris-setosa
4.900	3.100	1.500	0.100	id_10	Iris-setosa

Q&A

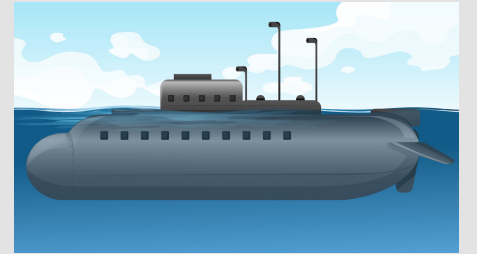
Dataset มาจากไหน ?

- Sample data (Rapidminer)
- UCI Machine Learning Repository
- Data World
- Kaggle
- Subreddit /Datasets
- R Package – DSLabs
- Real data

Classification

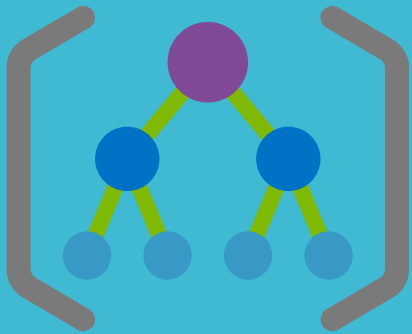
ตัวอย่างที่ 1

ทำนายผู้อยู่รอดจากเรือ Titanic

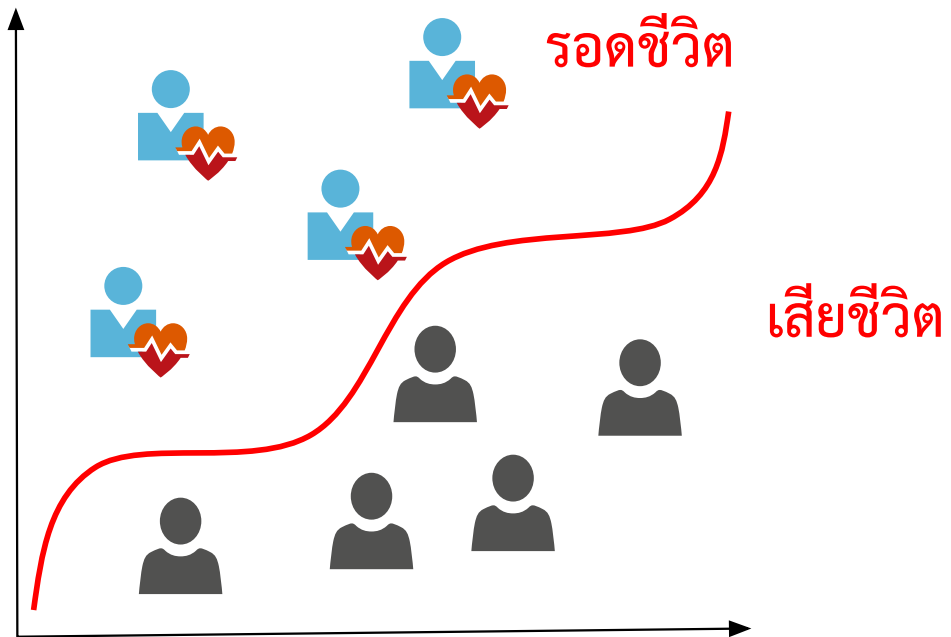


ตัวอย่าง

การจำแนก

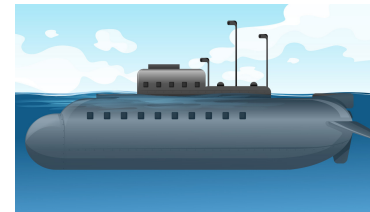


- ใช้ข้อมูลที่มี จำแนกกลุ่มประชากรออกจากกัน
- จากตัวอย่างข้อมูล Titanic



ทดสอบการจำแนก

- Titanic กับผู้อยู่รอด

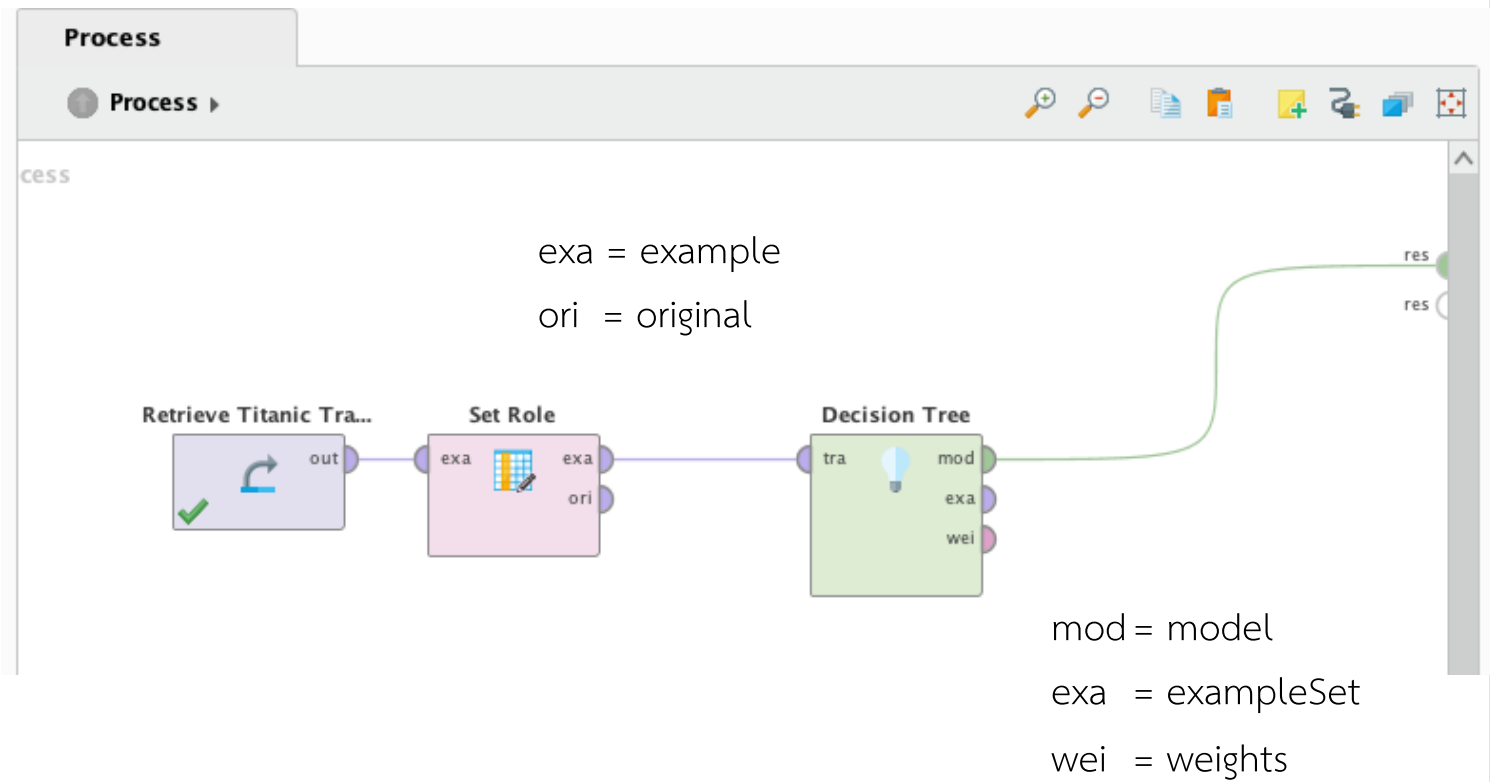


Label

Passenger Cl... Category	Name Category	Sex Category	Age Number	No of Sibling... Number	No of Parents o... Number	Ticket Number Category	Passenger Fare Number	Cabin Category	Port of Embarkation Category	Life Boat Category	Survived Category
First	Allen, Miss. E...	Female	29	0	0	24160	211.338	B5	Southampton	2	Yes
First	Allison, Mast...	Male	0.917	1	2	113781	151.550	C22 C26	Southampton	11	Yes
First	Allison, Miss....	Female	2	1	2	113781	151.550	C22 C26	Southampton	?	No
First	Allison, Mr. H...	Male	30	1	2	113781	151.550	C22 C26	Southampton	?	No
First	Allison, Mrs. ...	Female	25	1	2	113781	151.550	C22 C26	Southampton	?	No
First	Anderson, M...	Male	48	0	0	19952	26.550	E12	Southampton	3	Yes
First	Andrews, Mis...	Female	63	1	0	13502	77.958	D7	Southampton	10	Yes
First	Andrews, Mr....	Male	39	0	0	112050	0	A36	Southampton	?	No
First	Appleton, Mr...	Female	53	2	0	11769	51.479	C101	Southampton	D	Yes
First	Artagaveytia,...	Male	71	0	0	PC 17609	49.504	?	Cherbourg	?	No
First	Astor, Col. Jo...	Male	47	1	0	PC 17757	227.525	C62 C64	Cherbourg	?	No
First	Astor, Mrs. J...	Female	18	1	0	PC 17757	227.525	C62 C64	Cherbourg	4	Yes
First	Aubart, Mme...	Female	24	0	0	PC 17477	69.300	B35	Cherbourg	9	Yes
First	Barber, Miss....	Female	26	0	0	19877	78.850	?	Southampton	6	Yes
First	Barkworth. M...	Male	80	0	0	27042	30	A23	Southampton	B	Yes

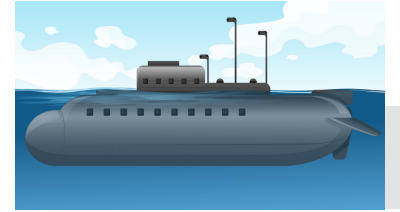
RapidMiner Studio

Design1

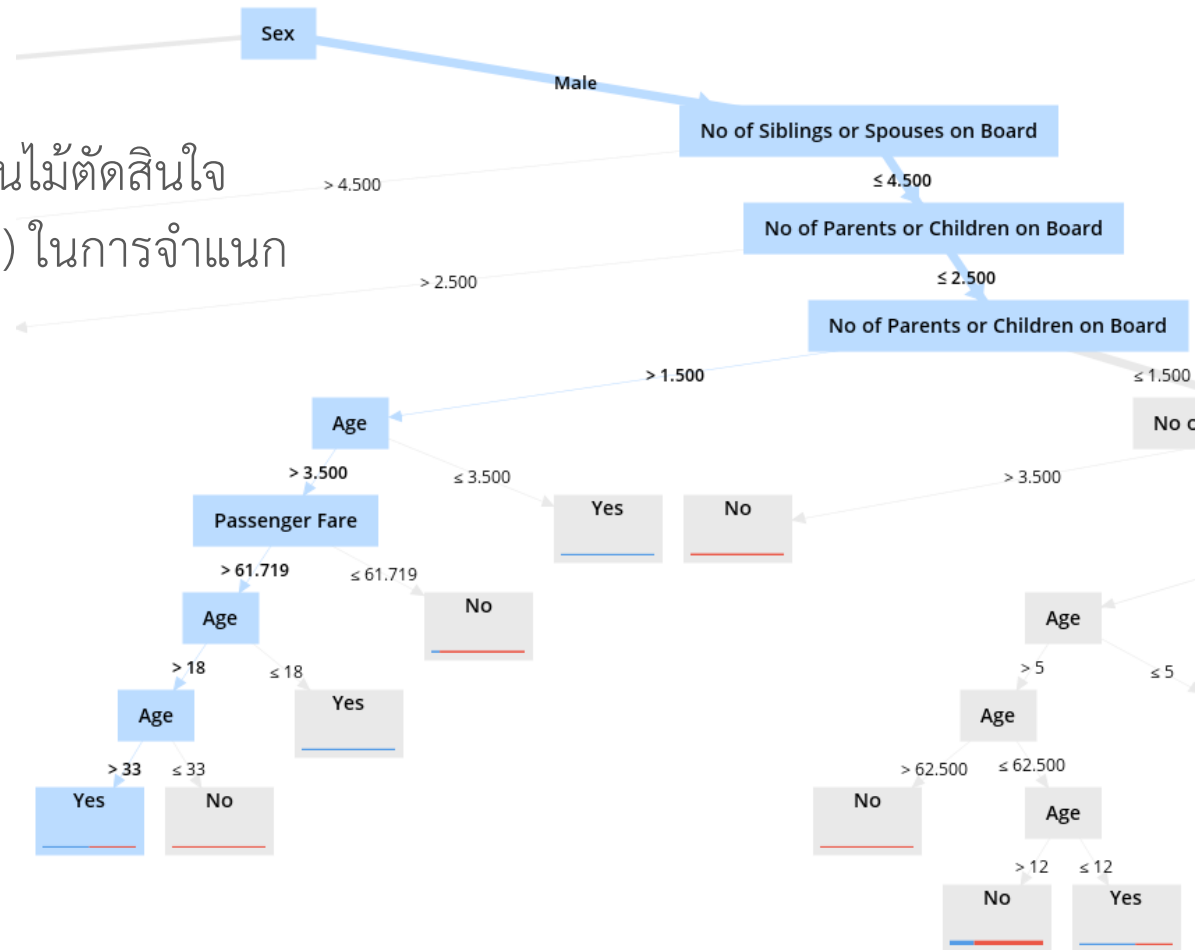


ทดสอบการจำแนก

- Titanic กับผู้รอด

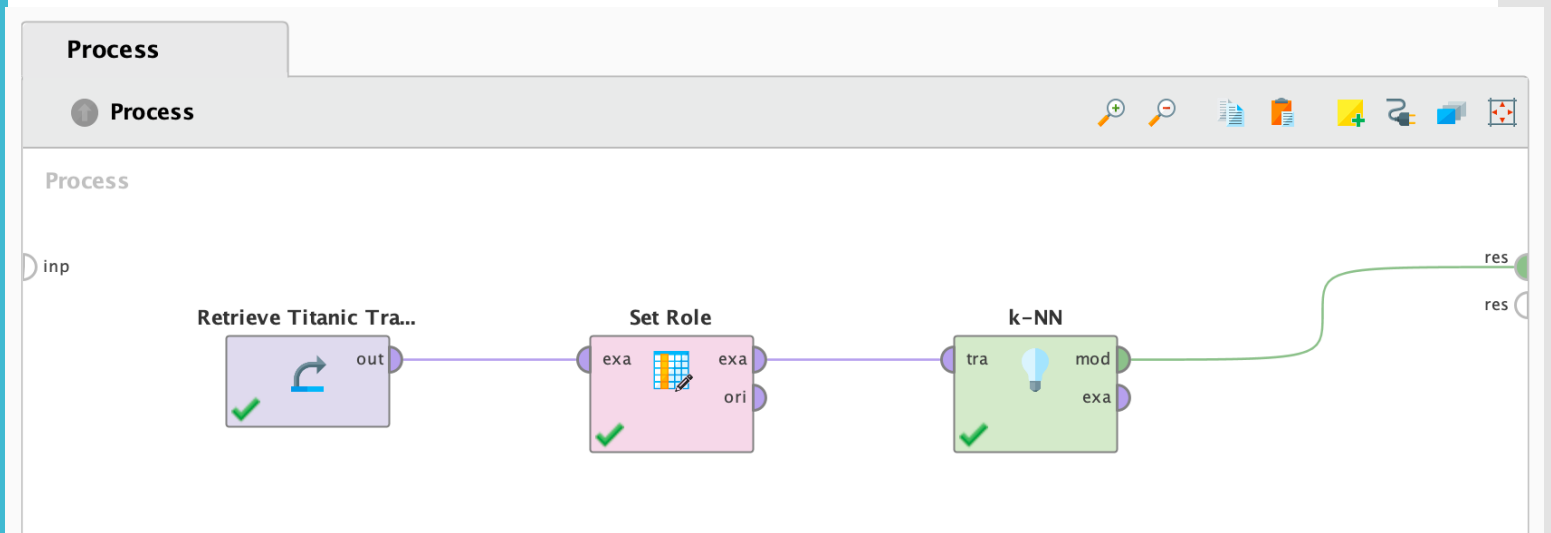


- ตัวอย่างการใช้ต้นไม้ตัดสินใจ (Decision Tree) ในการจำแนกข้อมูล Titanic



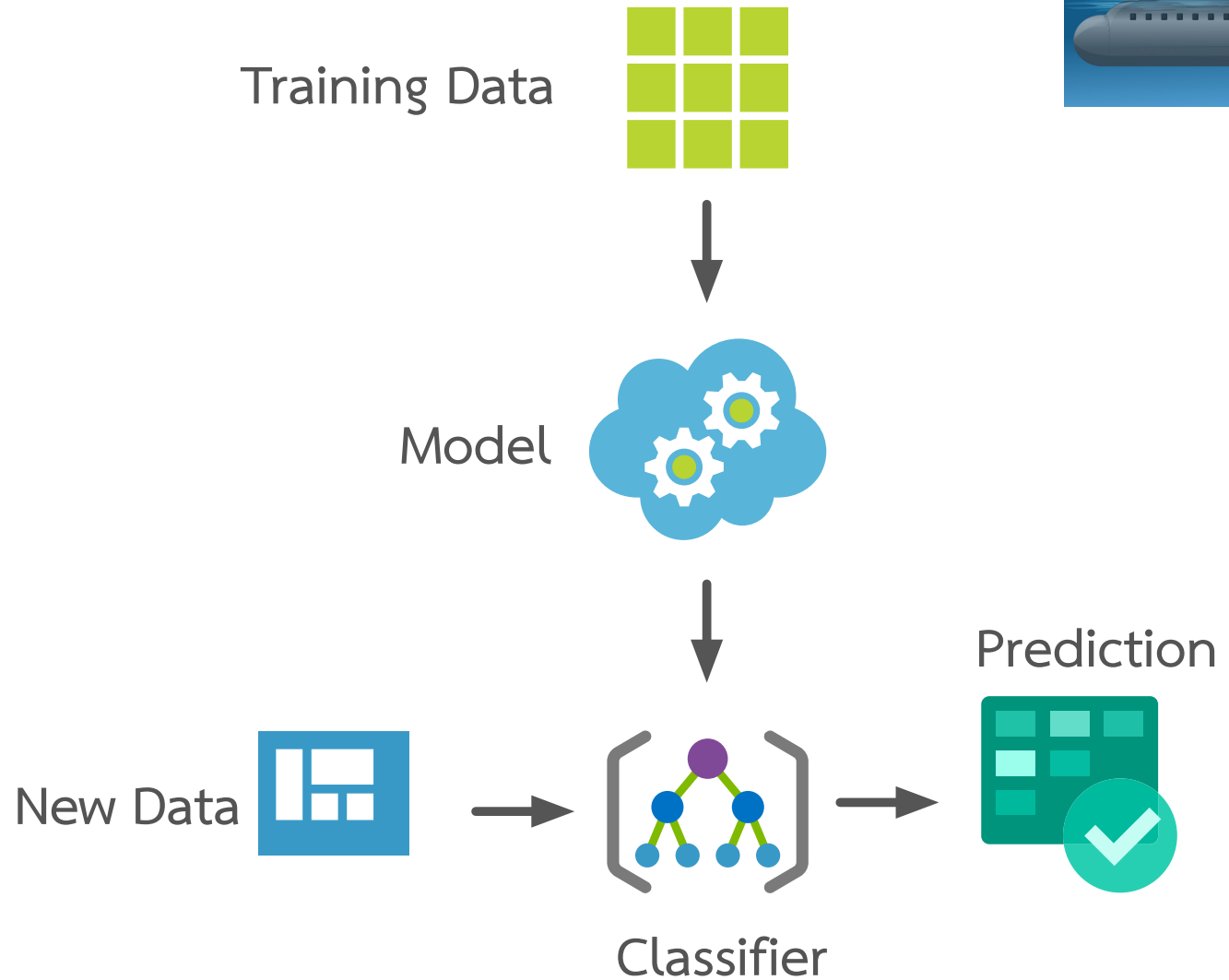
RapidMiner Studio

Design1
K-nn



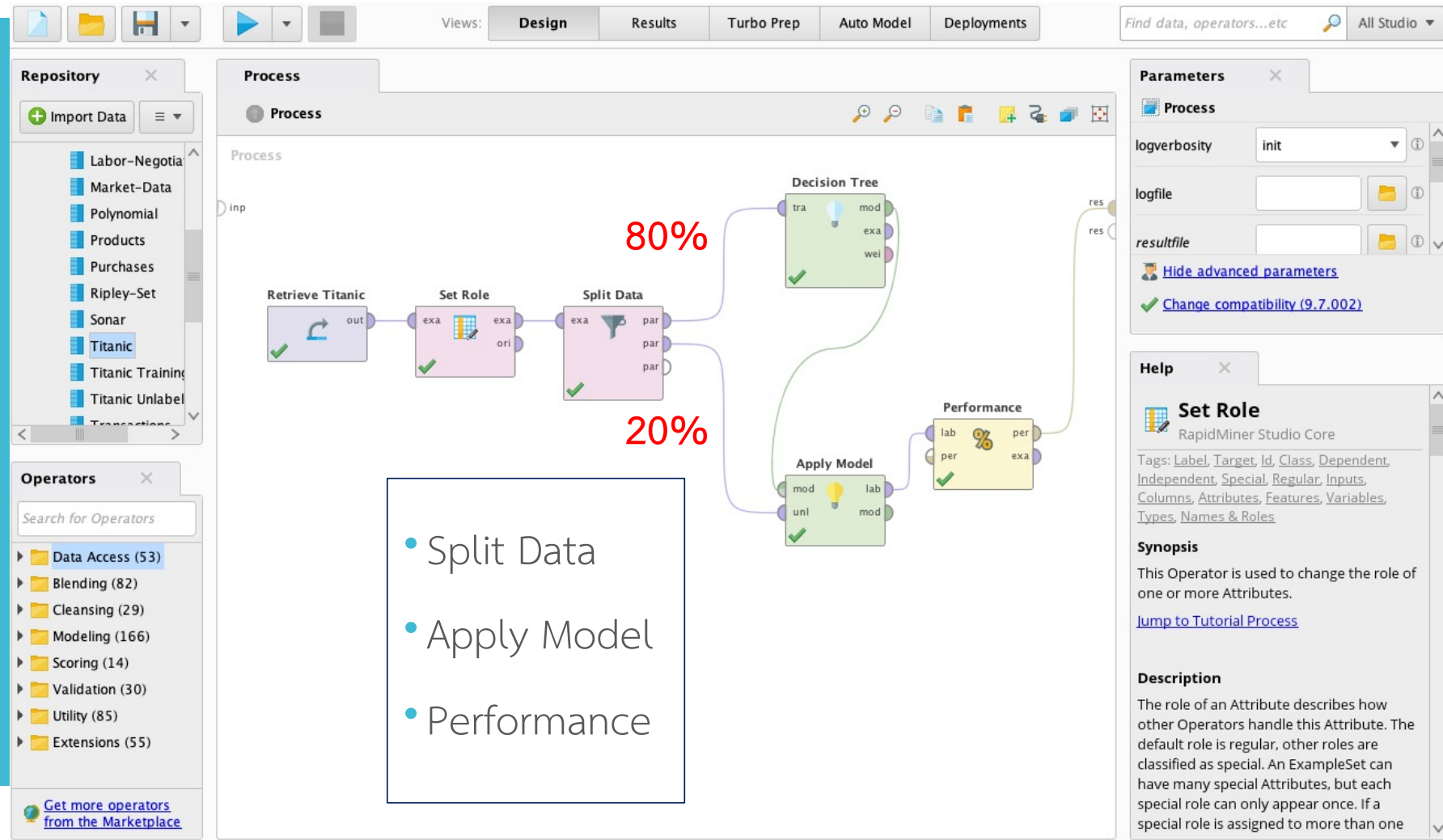
การประเมิน
ประสิทธิภาพ
แบบจำลอง

ตรวจว่าทำนายถูก
เท่าไร



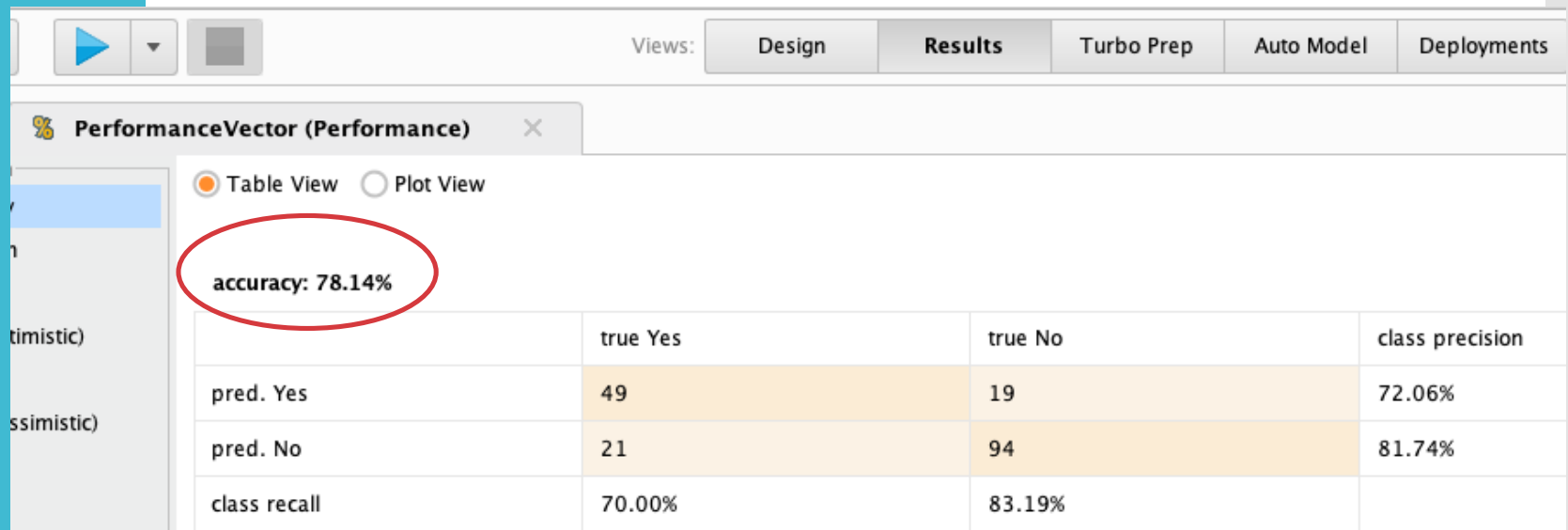
RapidMiner
Studio

Design2
Hole-out



RapidMiner Studio

Design2 Result



The screenshot shows the 'PerformanceVector (Performance)' window in RapidMiner Studio. The 'Table View' radio button is selected. A red circle highlights the text 'accuracy: 78.14%'. Below this is a confusion matrix table.

	true Yes	true No	class precision
pred. Yes	49	19	72.06%
pred. No	21	94	81.74%
class recall	70.00%	83.19%	

- ทำนายถูก (49+94) จาก sample ทั้งหมด (49+19+21+94)
คิดเป็น $143/183 = 78.14\%$

Q&A